ℓ_0 Neural Spike Train Inference via Tight Convexification

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- Final presentation for USC ISE-617 2026

Why analyze spike trains?

Contributions

Model

Optimization problem

$O(T^2)$ algorithm

Conclusion

- Neural spike train
- Why do we analyze them

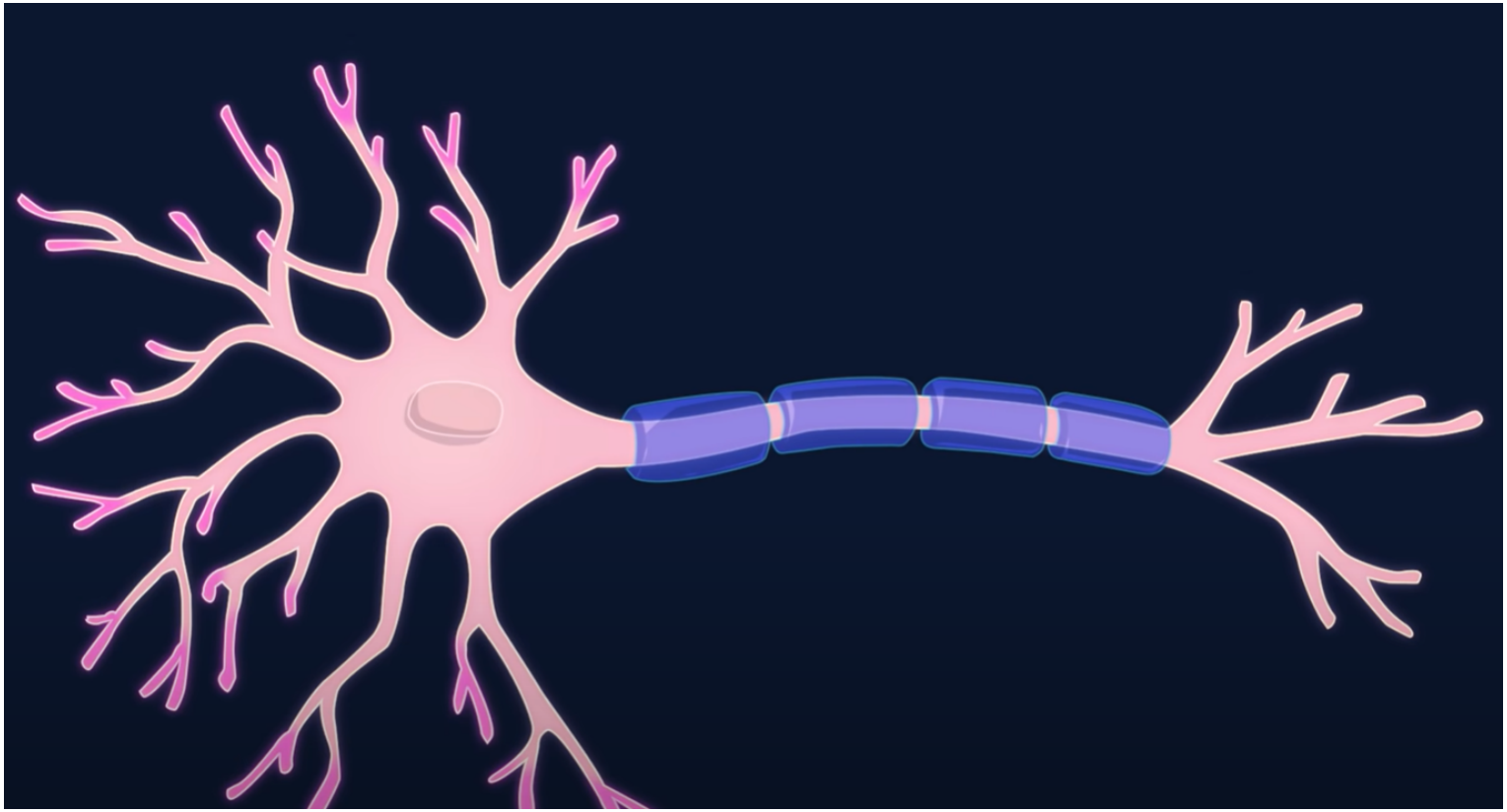

Figure: Neurons receive, process, and transmit information through a phenomenon called action potential

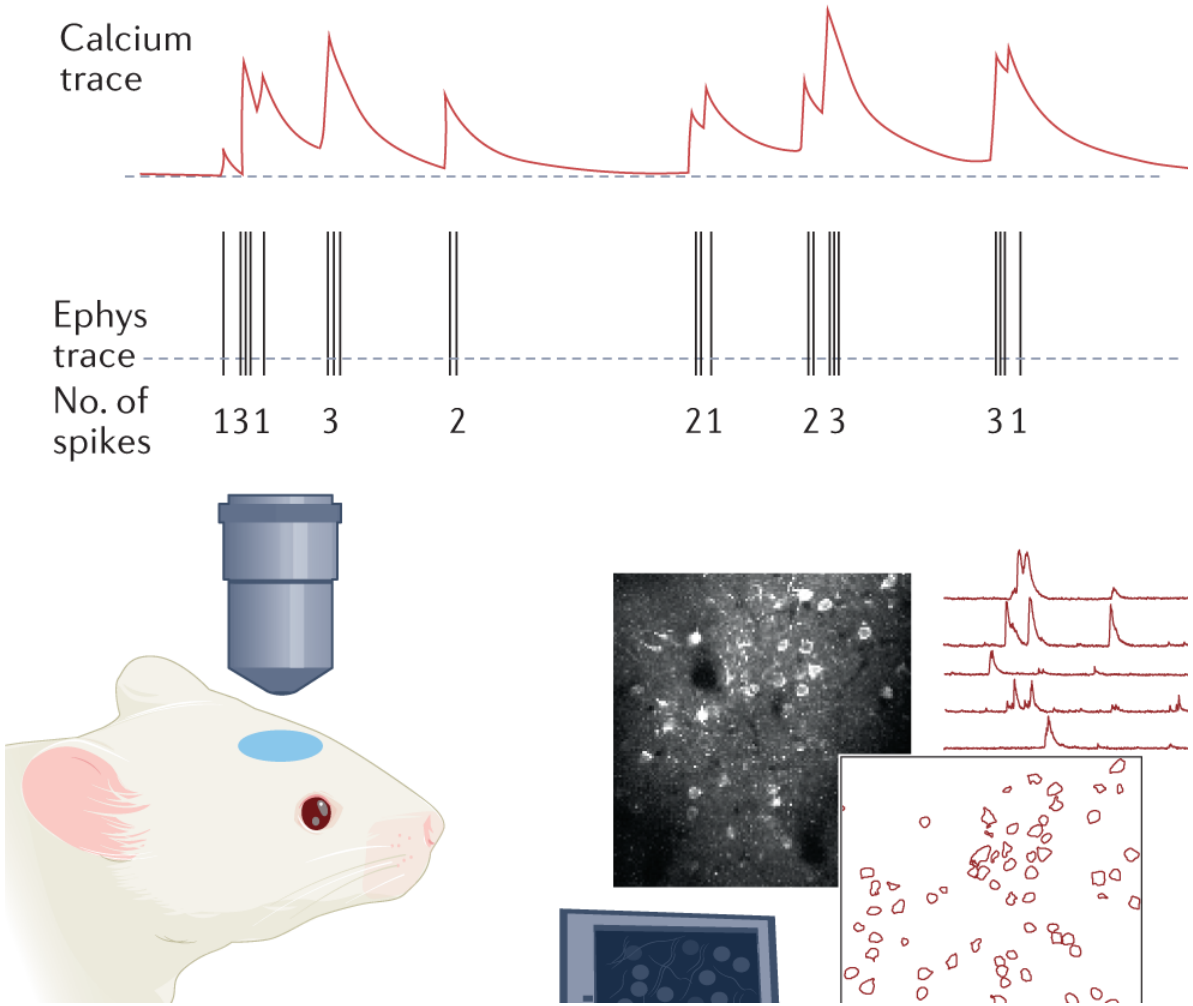

Figure: Calcium imaging records the action potentials - spike train.

Binary encoding

- In the spike train, the information is encoded in whether a neuron fires or not, rather than in the magnitude of the potential.
- ALL neural activities (your thoughts, orders for your action) are encoded this way.

Two research tasks

- Neural decoding: mind-reading
- Neural encoding: animal control

All require accurate spike train inference. Recordings are always noisy.

Two ways to improve

How can we better detect the spikes?

- Develop new recording technology
- Improve the detection algorithms (this paper)

Contributions

- Formulates the spike detection as ℓ_0 regularized optimization
- Transforms the problem in to change-point detection problem
- Devise an $O(T^2)$ solver based on dynamic programming
- Faster and more accurate than ℓ_1 regularization.

Three stages of models

- Spike sequence (fixed, unobserved)
- Underlying calcium concentration process (fixed, unobserved)
- Noise-added observations (random, observed)

1. Spike sequence

$s_t \geq 0,\ t1, \ldots, T$

- Example:
  $0,\ 3.5,\ 0,\ 0,\ 1.1,\ 0,\ 0,\ 2,\ \ldots$
- If $s_t > 0$: spike at time t
- If $s_t = 0$: no spike at time t
- Fixed unknown parameter of interest
- Sparsity assumption

2. Underlying calcium concentration

- First-order autoregressive process

$$C_t = \gamma C_{t-1} + s_t, \quad t = 2, 3, \ldots, T$$

- Spike appears abrupt, and gradually disappears.
- The effect of spikes can overlap
- Fixed unknown parameter

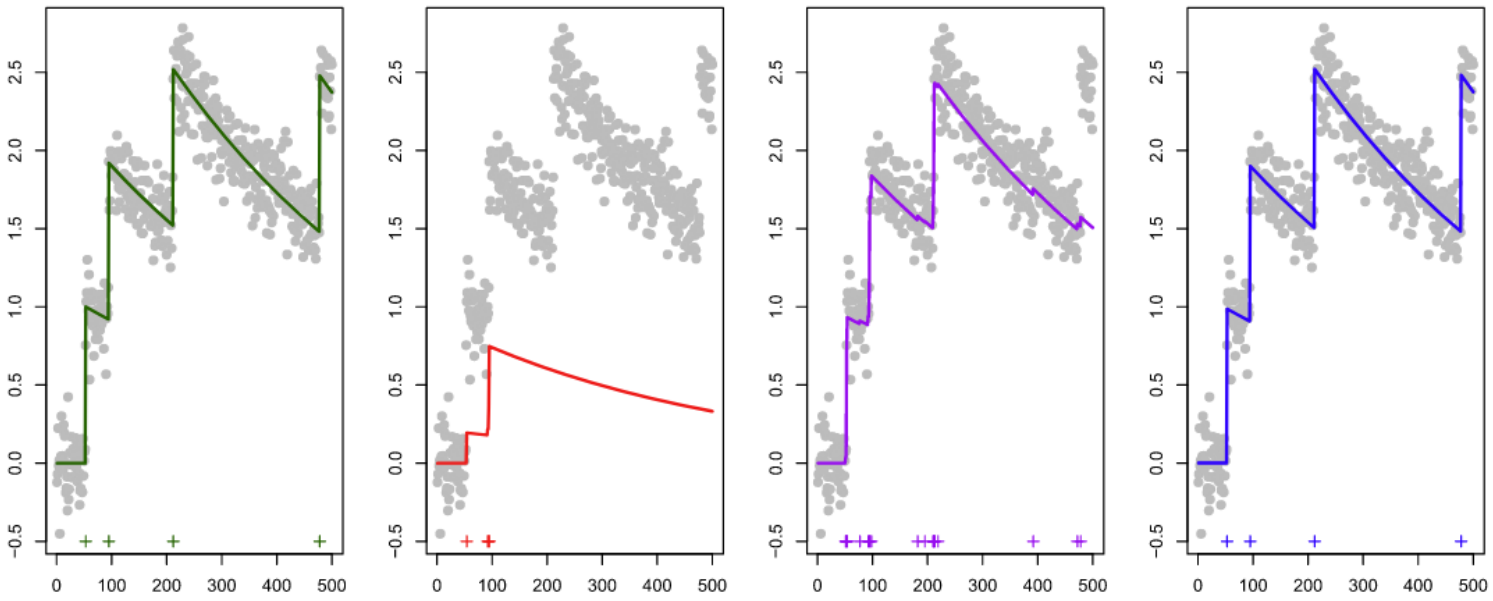

Figure: Spike appears abrupt, and gradually disappears

3. Observed fluorescence process

$$Y_t = c_t + \varepsilon_t, \quad \varepsilon_t \sim_{\text{ind}} (0, \sigma^2)$$

for $\quad t = 1, \ldots, T$

Optimization problem

$$\min_{c_1,\ldots,c_T,\, s_2,\ldots,s_T} \frac{1}{2}\sum_{t=1}^{T}(y_t - c_t)^2 + \lambda \sum_{t=2}^{T} \mathbb{1}(s_t = 0)$$

$$\text{s.t. } s_t = c_t - \gamma c_{t-1}.$$

- Regularization $\lambda \mathbb{1}(c_t \neq \gamma c_{t-1})$ ties c_t and c_{t+1}.
- This tie is active only when $c_t = \gamma C_{t-1}$ ($s_t = 0$) (no spike)

Another viewpoint

Thus, if the solution sets $s_t > 0$, than we actually solved the problem that decouples into $[1, t-1]$ and $[t+1, T]$, and t is a changepoint.

Thus, problem is equivalent to the changepoint detection problem:

$$(\text{spike magnitude} \longleftrightarrow \text{timepoint})$$

$$\begin{array}{ccccccccc} 0 & 1.1 & 0 & 0 & 3.5 & 0 & 2.33 & \cdots \\ & 2 & & & 5 & & 8 & \end{array}$$

Proposition 1

The problem is equivalent to

$$\min_{0=\tau_0<\tau_1<\cdots<\tau_k<\tau_{k+1}=T} \left\{ \sum_{j=0}^{k} \mathcal{D}(\tau_j+1, \tau_{j+1}) + \lambda k \right\}$$

for two timepoints $a = \tau_j + 1, \quad b = \tau_{j+1}$,

$$\mathcal{D}(a,b) := \arg \min_{c_a, c_{a+1}, \ldots, c_b} \frac{1}{2} \sum_{t=a}^{b} (y_t - c_t)^2$$

$$\text{s.t.} \quad c_t = \gamma c_{t-1}.$$

$$\mathcal{D}(a,b) := \arg\min_{c_a, c_{a+1}, \ldots, c_b} \frac{1}{2} \sum_{t=a}^{b} (y_t - c_t)^2$$

$$\text{s.t.} \quad c_t = \gamma c_{t-1}.$$

- Measures homogeneity of the measurement
- Has a closed-form formula (Proposition 2).

$$\min_{0=\tau_0<\tau_1<\cdots<\tau_k<\tau_{k+1}=T} \left\{ \sum_{j=0}^{k} \mathcal{D}(\tau_j + 1, \tau_{j+1}) + \lambda k \right\}$$

Find not too many timepoints such that between two consecutive timepoints, the measurements are homogeneous

Apply the dynamic programming algorithm proposed by Jackson et al. (2005, *IEEE Signal Processing Letters*) developed for changepoint detection.

$$F(T) := \min_{0=\tau_0<\tau_1<\cdots<\tau_k<\tau_{k+1}=T} \left\{ \underbrace{\sum_{j=0}^{k} \mathcal{D}(\tau_j+1, \tau_{j+1}) + \lambda k}_{\text{(A)}} \right\}$$

$$\text{(A)} = \sum_{j=0}^{k} [\mathcal{D}(\tau_j+1, \tau_{j+1}) + \lambda] - \lambda \quad (k+1 \text{ terms})$$

$$= \underbrace{\sum_{j=0}^{k-1} [\mathcal{D}(\tau_j+1, \tau_{j+1}) + \lambda] - \lambda}_{\tau_0, \ldots, \tau_k\text{: } k \text{ intervals, contributes to } F(\tau_k)} + \underbrace{\mathcal{D}(\tau_k+1, \tau_{k+1}) + \lambda}_{\tau_k+1, \tau_{k+1}}$$

Dynamic programming

Set $F(0) = -\lambda$

$$F(1) = \min_{\tau \in \{0\}} \{F(\tau) + \mathcal{D}(y_{\tau+1:1}) + \lambda\}$$

$$F(2) = \min_{\tau \in \{0,1\}} \{F(\tau) + \mathcal{D}(y_{\tau+1:2}) + \lambda\}$$

$\mathcal{D}$ calculation in constant time, at step t, reads t data: $O(\sum_{t=1}^{T} T) = O(T^2)$

- Formulates the spike detection as ℓ_0 regularized optimization
- Transforms the problem in to change-point detection problem
- Devise an $O(T^2)$ solver based on dynamic programming
- Faster and more accurate than ℓ_1 regularization (numerically).
- No statistical theory, just optimization